

Lab 9 - OpenClaw Cost Saving

Reduce AI running costs through smart model selection, context management, and free-tier options.

Lab environment: Hostinger VPS (from Lab 1) or Docker Desktop (Windows 10/11, macOS 12+, Ubuntu 22.04+)

Prerequisite: Lab 2 completed - a model provider configured. Estimated time: 20 minutes

Docker Desktop users: Prefix every openclaw command with docker exec -it openclaw.

Cost Overview

Approach	Cost	Notes
Claude Opus 4.8 API	High	Best quality, pay per token
Claude Sonnet 5 API	Medium	Good balance
Claude Haiku 4.5 API	Low	Fast, cheap, good for simple tasks
Groq Llama 3.3 70B	Free tier	No credit card, fast inference
Google Gemini 2.0 Flash	Free tier	Generous free quota
Ollama (local)	Zero API cost	Runs on your hardware

Strategy 1 - Switch to a Cheaper Model

The fastest cost saving is switching to a smaller, cheaper model.

```
# VPS
openclaw configure --section model

# Docker Desktop
docker exec -it openclaw openclaw configure --section model
```

In the wizard, select your provider and choose a cheaper model:

Switch	Cost Saving
Claude Opus → Haiku	~90% per token
GPT-4o → GPT-4o-mini	~85% per token
Any paid → Groq Llama	~100% (free tier)

Strategy 2 - Use Groq Free Tier for Testing

During learning and testing, Groq gives you fast inference for free.

```
# VPS
openclaw configure --section model

# Docker Desktop
docker exec -it openclaw openclaw configure --section model
```

In the wizard:

1. Select More... → Groq
2. Enter your gsk_... API key from <https://console.groq.com>
3. Select llama-3.3-70b-versatile

Switch back to a paid model for production when needed.

Strategy 3 - Use Ollama for Zero-Cost Local Testing

No API costs - the model runs on your machine.

Step 1 - Install Ollama:

```
# macOS / Linux
curl -fsSL https://ollama.com/install.sh | sh

# Windows - download from https://ollama.com/download
```

Step 2 - Pull a model: ``bash ollama pull llama3.2 ``

Step 3 - Configure OpenClaw: ``bash

VPS

openclaw configure --section model

Select Ollama → host: <http://localhost:11434> → model: llama3.2

Docker Desktop

docker exec -it openclaw openclaw configure --section model

Select Ollama → host: <http://host.docker.internal:11434> → model: llama3.2

****Step 4 (Docker Desktop only) - Register the model in config:****

The wizard sets the default model but does not register the model entry. Run this to fix it:

```
docker exec openclaw sh -c 'cat > /tmp/fix-ollama.js << "EOF" const fs=require("fs"); const
p=process.env.HOME+"/.openclaw/openclaw.json"; const c=JSON.parse(fs.readFileSync(p)); if(!c.models)
c.models={}; if(!c.models.providers) c.models.providers={}; if(!c.models.providers.ollama)
c.models.providers.ollama={api:"ollama",baseUrl:"http://host.docker.internal:11434",models:[]};
if(!c.models.providers.ollama.models) c.models.providers.ollama.models=[]; const
```

```
exists=c.models.providers.ollama.models.find(function(m){return m.id=="llama3.2";}); if(!exists)
c.models.providers.ollama.models.push({id:"llama3.2",name:"llama3.2"});
c.models.providers.ollama.baseUrl="http://host.docker.internal:11434"; fs.writeFileSync(p,JSON.stringify(c,null,2));
console.log("done"); EOF' MSYS_NO_PATHCONV=1 docker exec openclaw node /tmp/fix-ollama.js ``
```

Expected output: done

Step 5 - Set as default and restart: ``bash docker exec -it openclaw openclaw models set ollama/llama3.2 docker restart openclaw ``

Strategy 4 - Context Compaction

Context compaction summarises old conversation history to reduce tokens sent per request. This can cut costs by 50-70%.

```
# VPS
openclaw config set context.compaction enabled
openclaw config get context.compaction

# Docker Desktop
docker exec -it openclaw openclaw config set context.compaction enabled
docker exec -it openclaw openclaw config get context.compaction
```

Expected: `` context.compaction: enabled ``

Strategy 5 - Use Google Gemini Free Tier

Google Gemini 2.0 Flash has a generous free quota with no credit card needed for the free tier.

```
# VPS
openclaw configure --section model

# Docker Desktop
docker exec -it openclaw openclaw configure --section model
```

In the wizard:

1. Select Google
2. Enter your Alza... key from <https://aistudio.google.com/app/apikey>
3. Select gemini-2.0-flash

Cost Saving Summary

Action	Potential Saving
Claude Opus → Haiku	~90% per token
Switch to Groq free tier	100% during testing
Switch to Ollama locally	100% API cost
Enable context compaction	50-70% per session
Google Gemini free tier	100% within free quota

Verification

Check	Expected
openclaw configure --section model	Wizard opens, model changeable
openclaw config get context.compaction	enabled
Chat test after model switch	Agent replies with new model

Troubleshooting

Symptom	Fix
After switching model, agent stops responding	Re-run openclaw configure --section model and re-enter credentials
Ollama unreachable from Docker	Use http://host.docker.internal:11434 not localhost
Unknown model: ollama/llama3.2 error in logs	Run Step 4 above - the model must be registered in models.providers.ollama.models[]
Git Bash ! expansion error in script	Use MSYS_NO_PATHCONV=1 prefix before docker exec commands with Linux paths
Context compaction causes incomplete answers	Increase threshold: openclaw config set context.compaction-threshold 100000
Groq rate limits	Free tier has rate limits - wait 1 minute and retry

Reference

- Model providers: <https://docs.openclaw.ai/providers>
- Groq (free): <https://console.groq.com>
- Google AI Studio: <https://aistudio.google.com>
- Ollama: <https://ollama.com>